

week-3-lesson-2

Example 1

Calculate the sum of the alternating series $S = 1 - 2 + 3 - 4 + 5 - 6$ up to $N = 6$ using a single for loop. You cannot use any conditional checks. You must update a sign multiplier variable directly inside the loop's update field.

Example 2

Print a multiplication table for a 3 times 3 grid, but only calculate and print values where the row index is greater than or equal to the column index (e.g., lower-triangular). You must do this without checking if the indices are equal; control the execution strictly through the inner loop's boundary condition.

Example 3

Euclid's Branchless GCD & LCM Concept: A while loop repetitively executes as long as its condition evaluates to true. Euclid's Greatest Common Divisor (GCD) algorithm runs until the remainder variable hits 0.

Calculate both the GCD and the Least Common Multiple (LCM) of two integers, $u = 150$ and $v = 35$, using a while loop. You must not use any conditional statements. $LCM = (u \times v) / GCD$

Example 4

Pure Mathematical Decimal-to-Octal Converter Concept: Digits of a number can be isolated and extracted using mathematical division and modulus operators inside a while loop.

Convert a decimal integer (e.g., 83) into its octal representation and store the result as a decimal-formatted integer (e.g., 123) using a while loop. You cannot use arrays, strings, or printf base formatters to do the conversion.

Example 5

Write a program that counts the number of digits in an integer using a do while loop. Ensure that value 0 is correctly evaluated as having exactly 1 digit without using an if statement.

Example 6

Write a program that counts the number of digits in an integer using a do while loop. Ensure that negative value is correctly evaluated without using an if statement.